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RESEARCH ARTICLE

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Efficacy of probiotics pretreatment in *Helicobacter pylori* eradication therapy: a systematic review and meta-analysis of clinical outcomes

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ABSTRACT

Background: Probiotics inhibit *Helicobacter pylori (H. pylori)* growth and alter gut microbiome, in addition to alleviating the side effects of *H. pylori* eradication. It has also been reported that *H. pylori* be eradicated after fecal microbiota transplantation in some cases. However, whether probiotics used before *H. pylori* eradication improves the eradication rate remains unclear. This study evaluates their role through a systematic review and meta-analysis.

Methods: We searched PubMed, EMBASE, the Cochrane Library, and the Conference Proceedings Citation Index up to January 31, 2024, to identify randomized controlled studies (RCTs) assessing the efficacy of probiotics used before *H. pylori* eradication. Meta-analyses of eradication rates were performed.

Results: Twelve eligible RCTs with 2,144 participants were included. The intention-to-treat analysis revealed that the overall eradication rate of *H. pylori* was higher in the probiotics pretreatment group compared to the control group (80.34% vs. 70.49%), with a risk ratio (RR) of 1.14 (95% CI: 1.08 to 1.19; I^2 = 36%) and side effects were less (16.0% vs. 28.3%, RR = 0.59, 95% CI 0.41 to 0.84). The per-protocol analysis yielded similar results (86.43% vs. 76.88%, RR = 1.12, 95% CI: 1.08 to 1.17; I^2 = 57%). Subgroup analyses, considering factors like geographic location, eradication regimens, and probiotic combinations, consistently confirmed the benefits. Finally, probiotics pretreatment durations of 14 days or more, as well as the study designs *pre vs. free* and *pre and combine vs. free*, demonstrated significant effects. Shorter durations and other study designs with fewer studies did not show significant effects.

Conclusion: There is moderate to high evidence to suggest that probiotics pretreatment improves *H. pylori* eradication rate and reduces side effects. These findings highlight the potential value of gut microbiome modulation in *H. pylori* treatment and offer a new direction for addressing the challenges of antibiotic resistance and treatment failure.

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Helicobacter pylori; pretreatment; probiotics; microbiome; meta-analysis

Introduction

The global prevalence of *Helicobacter pylori* (*H. pylori*) infection remains above 40% [1]. *H. pylori* infection, which invariably causes gastritis, is difficult to clear naturally unless treated with eradication therapies consisting of antibiotics and acid suppressants [2]. Persistent infection can lead to chronic atrophic gastritis, peptic ulcers, gastric cancer, gastric mucosa-associated lymphoid tissue lymphoma etc. *H. pylori* has

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also been linked to an increased risk of colorectal cancer and Alzheimer's disease [3,4]. However, the rise in antibiotic resistance, fueled by widespread antibiotic use, has caused *H. pylori* eradication rate to decline [5]. Additionally, *H. pylori* eradication therapies also induce an increase in long-term resistant bacteria communities in the gut [6].

Probiotics, live microorganisms that bring health benefits, have emerged as a potential adjunct to *H. pylori* therapy. Randomized controlled trials (RCTs) and meta-analyses suggest that probiotics can reduce side effects associated with *H. pylori* eradication treatments, though their impact on *H. pylori* eradication rate remains unclear [7–14]. Emerging evidence highlights the gut microbiome's role in *H. pylori* clearance. Probiotics may inhibit *H. pylori* growth through producing antibacterial substances, competing for ecological niches, and modulating the immune system [15–18]. Probiotics also modulate the microbiome across the digestive tract [19–22], potentially altering the environment in which *H. pylori* thrives and leading to the decolonization of pathogenic bacteria [23]. Furthermore, faecal microbiota transplantation (FMT) has even been reported to clear *H. pylori* in some cases, underscoring the microbiome's importance [24].

Despite these findings, it remains unclear whether using probiotics before *H. pylori* eradication therapy improves outcomes. This study aims to address this gap by evaluating the efficacy of probiotics administered prior to *H. pylori* treatment through a systematic review and meta-analysis.

Methodology

Registration of review protocol

This systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline [25]. The study protocol was registered on the International Prospective Register of Systematic Reviews (CRD42021233382; 22 Feb 2021; Updated 07 Jan 2024).

Search strategy

A comprehensive literature search of PubMed, EMBASE, the Cochrane Library, and the Conference Proceedings Citation Index was carried out up till January 2024 with the following search strategy (*Helicobacter pylori* OR *H. pylori* OR *Campylobacter Pylori*) AND (probiotic OR probiotics OR yeast OR yeasts OR yogurt OR *Lactobacillus* OR *Bifidobacterium* OR *Saccharomyces* OR *Streptococcus* OR *Enterococcus* OR *Bacillus* OR prebiotics OR *Lactococcus* OR fermented milk OR *Clostridium butyricum* OR *Bacillus subtilis* OR *Bacillus licheniformis* OR Curd). We also manually checked the reference lists of RCTs and systematic reviews identified by previous searches to include other potentially eligible studies.

Eligibility criteria

The title and abstract were used to assess the eligibility of the records. Eligible records were screened for duplicates. Data were extracted from these eligible articles independently by two authors (TM and SX), and any disagreements were resolved through discussion with a third reviewer (ZY). RCTs that compared the pretreatment of probiotics for *H. pylori* eradication (pretreatment group) with current mainstream guideline-recommended therapies (control group) [7,26], with or without the use of probiotics, were included. Reviews, meta-analyses, comments, letters, case reports, case series, animal studies, and studies without controls were excluded. Additionally, studies that did not report the *H. pylori* eradication rate, used serological tests to diagnose active *H. pylori* infection, had an interval of less than 4 weeks for *H. pylori* re-check, or did not report the interval were also excluded.

Risk of bias assessment

Two investigators (TM and ZY) separately rated the risk of bias in the included RCTs according to the risk of bias assessment tool (RevMan 5.4, Nordic Cochrane Center, Cochrane Collaboration, Copenhagen, Denmark).

The assessment covered the following bias domains: random sequence generation (selection bias), allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), incomplete outcome data (attrition bias), selective reporting (reporting bias), and others. Each domain was categorized as having a low, high, or uncertain risk of bias. If RCTs ≥ 10 was included in the meta-analysis, funnel plots were constructed to assess the risk of publication bias.

Data extraction

We extracted the following information from the included studies: (a) year of publication, region, first author, study design, participants' characteristics, number of patients in each treatment arm, methods used to confirm *H. pylori* infection; (b) the type, dose, and duration of probiotics used; (c) initial or rescue therapy, regimen of eradication, eradication rates, side effects, number of patients in whom *H. pylori* infection was successfully eradicated (either provided directly or calculated from the intention-to-treat [ITT] and the per-protocol [PP] analyses).

Data synthesis and analysis

The endpoints of interest in this meta-analysis were the *H. pylori* eradication rate. To evaluate the heterogeneity among the included studies, the inconsistency index (I^2) statistic was performed. An I^2 value larger than 50% indicated a high degree of heterogeneity, and a random-effects model was used to pool studies for analyses. Otherwise, a fixed-effects model was used [27]. Statistical heterogeneity was assessed using sensitivity analyses to determine whether any clinical heterogeneity contributed to such a statistical difference.

Both ITT and PP analyses were used to determine the clinical outcomes. Subgroup analyses were performed to compare the *H. pylori* eradication rates across studies from different countries, eradication regimens, *H. pylori* group, probiotic types (single probiotic strains or combinations of probiotics), durations of probiotics used before *H. pylori* eradication therapies, and study designs of probiotics pretreatment. We summed dichotomous outcome measures as the risk ratio (RR) along with 95% confidence interval (CI) using RevMan 5.4. The results of this meta-analysis were reported as forest plots.

Results

Study selection

A flow diagram of this study is presented in Figure 1. Initially, 867 articles were identified through the primary literature search. Of these, 605 were excluded due to being duplicates, irrelevant to the analysis, or not RCTs. A total of 262 articles were retrieved for further review.

Eight studies were excluded because they were either incomplete or unpublished after registration. 153 articles were excluded as they did not involve the use of probiotics prior to *H. pylori* eradication treatment. Further exclusions included 52 articles that lacked current mainstream guideline-recommended therapies as controls, 29 articles that did not involve probiotics, and two similar studies published by authors from the same institution within a year. Two articles were excluded for not reporting the *H. pylori* eradication rate, and two others were excluded because the *H. pylori* re-check interval was either less than four weeks or not reported. One article was excluded as it was a conference report, and the final report of the same study had already been published. Another article was excluded for using non-viable probiotics. Finally, twelve studies involving 2,144 patients met the inclusion criteria and were included in the meta-analysis [28–39].

Characteristics of eligible studies

The characteristics of the included studies are summarized in Table 1. All studies involved adults, with no pediatric participants. Six studies from China included 1,156 patients, while six studies from Japan, the United States, Italy, Iraq, Iran, and the UAE included 988 patients. The studies covered both initial

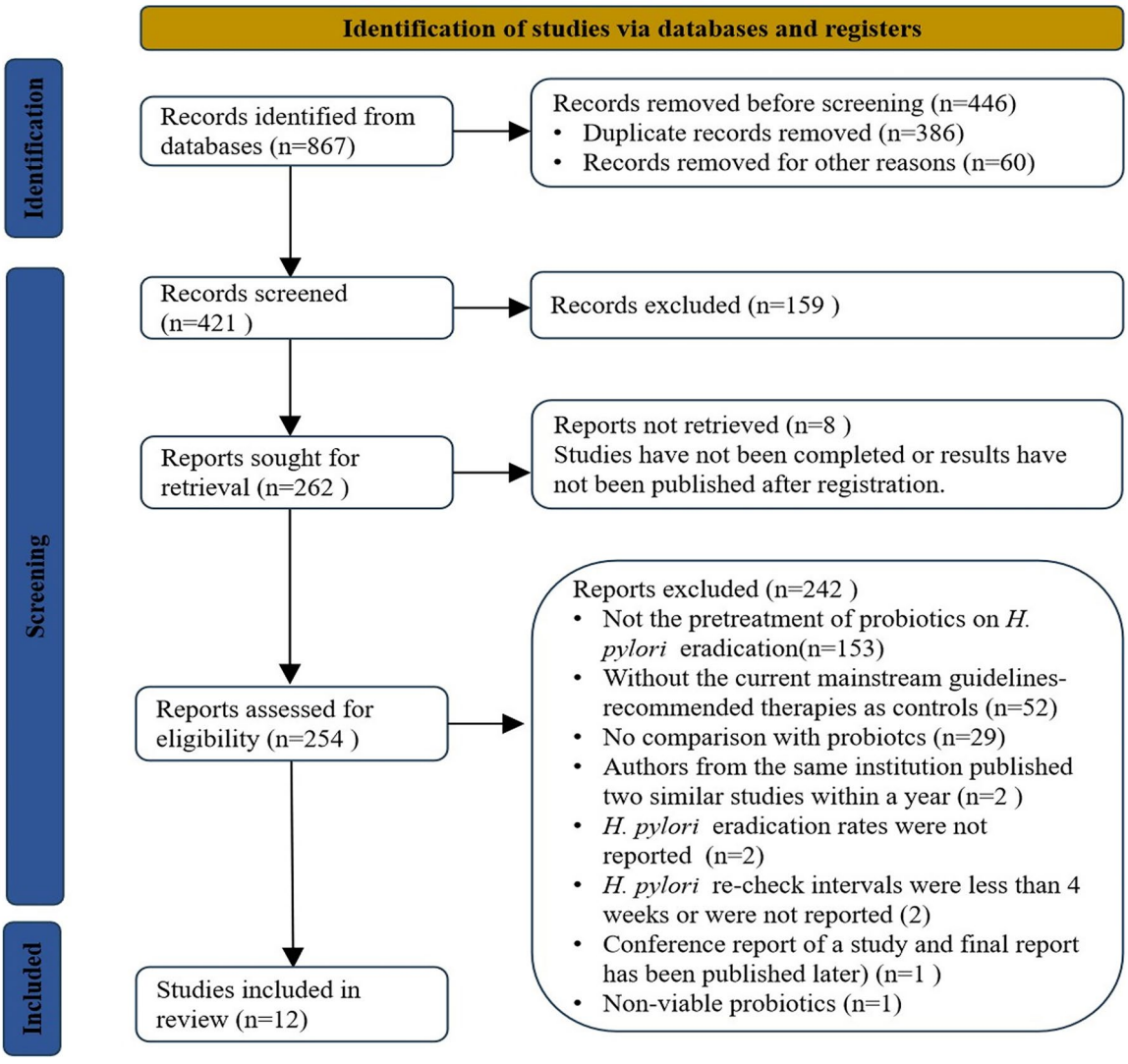


Figure 1. Preferred Reporting Items for systematic Reviews and meta-analyses flow diagram of study selection.

and rescue therapies for *H. pylori* infection, with four studies not specifying the therapy type and one including both. Six studies used standard triple therapy, and six used bismuth quadruple therapy. Nine studies used a combination of probiotics, while three used a single probiotic. Probiotic duration varied: 3-7 days (3 studies), 14 days (6 studies), and 21 days or longer (3 studies). The most common pretreatment comparisons were *pre vs. free* (7 studies) and *pre and combine vs. free* (4 studies), with fewer studies exploring other variations.

Outcomes of overall eradication rate

The eradication rate of *H. pylori* was evaluated across twelve studies involving 2,144 patients. By ITT analysis, the pooled eradication rate was 80.34% (707/880) in the pretreatment group and 70.49% (891/1,264) in the control group. Pretreatment with probiotics significantly improved the eradication rate (RR = 1.14, 95% CI 1.08 to 1.19, $p < 0.001$), with low heterogeneity observed between studies (I^2 = 36%) (Figure 2). Similarly, PP analysis revealed a pooled eradication rate of 86.43% (707/818) in the pretreatment group and 76.88% (891/1,159) in the control group. Probiotics pretreatment also improved the eradication rate (RR = 1.12, 95% CI 1.08 to 1.17, $p < 0.001$), although heterogeneity was higher (I^2 = 57%) (Figure 3).

Table 1. The characteristics of included studies.

Study	Subgroup(n)	Male/ Female	Age (years)	Hp infection. Initial diagnosis / Re-check	Hp group	Regimen of eradication	Probiotics used before Hp eradication treatment / Dose / Duration	Eradication rate (ITT/ PP, %)
Sheu (2006, China) [28]	Pre (69) Free (69)	37/31 38/30	48.9 46.4	UBT / UBT	Rescue	BQT	L. acidophilus La5, Bifidobacterium lactis Bb12, Lactobacillus bulgaricus, Streptococcus thermophilus / 2×10^{11} CFU / 14d	85.5/90.8 71.0/76.6
Deguchi (2012, Japan) [29]	Pre + combine (115) Free (114)	76/39 66/48	55.9 57.8	Culture, RUT, histology / UBT, SAT, culture	Initial	STT	Lactobacillus gasseri/ $\geq10^9$ / 21 d	82.6/85.6 69.3/74.5
Du (2012, China) [30]	Pre (78) Free (79) After (77)	32/44 33/45 26/48	44.9 ± 13.8 48.0 ± 13.3 48.2 ± 12.2	UBT, RUT, histology / UBT	Initial	STT	Acidophilus, Streptococcus faecalis, Bacillus subtilis /$\geq4.5\times10^7$/ 7d	79.5/81.6 60.8/61.5 79.2/82.4
Dajani (2013, United Arab) [31]	Pre + combine (95) Free (106) Combine (100)	47/48 51/55 51/49	19-61 17-64 17-62	RUT, histology, UBT, SAT / UBT	Initial or rescue	STT	*Bifidobacterium infantis* / 6×10^9 CFU/ 14d	90.5/90.5 68.9/68.9 83.0/83.0
Francavilla (2014, United States) [32]	Pre (50) Free (50)	18/32 21/29	22-65 18-61	UBT / UBT	Not report	STT	L. reuteri DSM 17938 and ATCC PTA 6475 /2×10^8 CFU / 28d	75.0/76.7 65.9/67.4
Wang F (2014, China) [33]	Pre (90) Free (90)	66/24 68/22	42 ± 14 40 ± 14	UBT, histology / UBT	Rescue	BQT	Lactobacillus acidophilus, Streptococcus faecalis and Bacillus subtilis / $>3\times10^7$ CFU / 14d	66.7/81.1 75.6/78.2
Peng (2016, China) [34]	Pre (48) Free (48) Combine (48)	27/21 38/18 31/17	42.7 ± 15.0 47.4 ± 16.7 45.0 ± 15.7	UBT / UBT	Rescue	STT	Bifidobacterium, Lactobacillus bulgaricus and Streptococcus thermophilus / $>6\times10^7$ CFU / 14d	81.3/83.0 60.4/69.0 83.3/83.3
Shafaghi (2016, Iran) [35]	Pre + combine (38) Combine (38)	18/20 17/21	43.7 ± 13.3 43.3 ± 12.2	Histology / UBT	Not report	BQT	Lactobacillus, Bifidobacterium, Streptococcus thermophilus /4×10^8 CFU/ 3d	92.1/92.1 63.2/100
Jin (2018, China) [36]	Pre (70) Pre + combine (70) Free (70) Combine (70)	37/33 41/29 43/27 39/31	18-70	UBT / UBT	Rescue	BQT	Bifidobacterium, Lactobacillus bulgaricus and Streptococcus thermophilus / $>6\times10^7$ CFU / 28d	75.7/84.1 77.1/84.4 62.9/72.1 71.4/79.4
Wang L (2018, China) [37]	Pre (60) Free (60) Combine (60)	34/26 36/24 33/27	40.3 ± 11.5 39.5 ± 10.4 38.6 ± 12.3	UBT / UBT	Rescue	BQT	Lactobacillus acidophilus, Streptococcus faecalis and Bacillus subtilis / >3×107 CFU / 14d	91.7/91.7 65.0/65.0 80.0/80.0
Ameen (2019, Iraq) [38]	Pre + combine (37) Free (31) Combine (38) After (24)	Not report	17-70	SAT and UBT / SAT or UBT	Not report	STT	lactobacillus acidophilus, lactobacillus salivarius, bifidobacterium bifidum and streptococcus thermophillus / 1×10^9 CFU/ 14d	45.9/89.5 41.9/76.5 47.4/78.3 62.5/83.3
Marinelli (2022, Italy) [39]	Pre + combine + after (66) Free (98)	23/43 20/78	50.0–59.3 53.9–62.0	UBT, histology, / UBT	Not report	BQT	Lactobacillus rhamnosus GG / 6×10^7 CFU / 7d	74.2/94.2 83.7/96.5

BQT: bismuth quadruple therapy; CFU: colony forming unit; Hp: *Helicobacter pylori*; ITT: intention-to-treat; PP: per-protocol; Pre: pretreatment; RUT: rapid urease test; SAT: stool antigen test; STT: standard triple therapy; UBT: urea Breath Test.

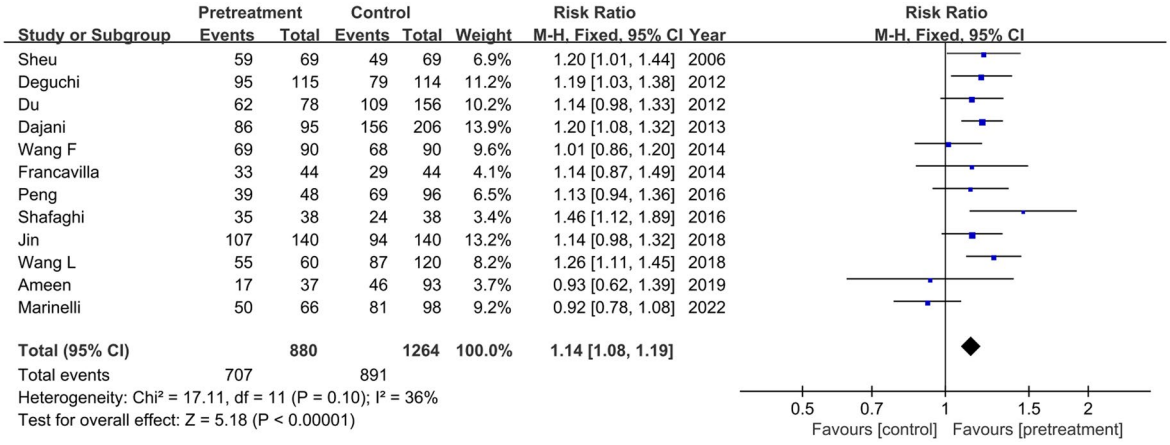


Figure 2. Forest Plot of randomized controlled trials of *H. pylori* eradication therapy: Efficacy on *H. pylori* eradication rate (intention-to-treat analysis).

Subgroup analysis for H. pylori eradication rate by region

In ITT analysis, studies from China (6 studies) showed a higher eradication rate in the pretreatment group (RR = 1.14, 95% CI 1.07 to 1.22, $p < 0.001$, $I^2 = 0\%$), while studies outside China (6 studies) also demonstrated higher rates (RR = 1.13, 95% CI 1.05 to 1.22, $p < 0.001$, $I^2 = 60\%$). In PP analysis, studies from China reported similar results (RR = 1.13, 95% CI 1.07 to 1.20, $p < 0.001$, $I^2 = 0\%$), whereas studies outside China showed higher rates (RR = 1.11, 95% CI 1.05 to 1.18, $p < 0.001$, $I^2 = 74\%$). Both subgroups had consistent findings, with lower heterogeneity in studies from China (Table 2, Supplement Figures S1–S2).

Subgroup analysis for H. pylori eradication rate by eradication regimen

In ITT analysis, standard triple therapy (6 studies) showed a higher eradication rate in the pretreatment group (RR = 1.15, 95% CI: 1.07–1.23; $I^2 = 0\%$), while the bismuth quadruple therapy (6 studies) also demonstrated higher rates (RR = 1.13, 95% CI: 1.05–1.21; $I^2 = 66\%$). In PP analysis, standard triple therapy yielded similar results (RR = 1.15, 95% CI: 1.08–1.22; $I^2 = 0\%$), whereas bismuth quadruple therapy showed slightly lower rates (RR = 1.10, 95% CI: 1.04–1.16; $I^2 = 75\%$). Both regimens were effective (Table 2, Supplement Figures S3–S4).

Subgroup analysis for H. pylori eradication rate by H. pylori group

In ITT analysis, consistent results were observed across different *H. pylori* groups (initial therapy, rescue therapy, and unclear). The initial *H. pylori* group (2 studies) exhibited a higher eradication rate in the pretreatment group (RR = 1.17, 95% CI: 1.05 to 1.30; $I^2 = 0\%$). The rescue *H. pylori* group (5 studies) also showed increased eradication rates (RR = 1.14, 95% CI: 1.06 to 1.23; $I^2 = 12\%$). Similarly, the unclear *H. pylori* group (5 studies) demonstrated higher efficacy (RR = 1.11, 95% CI: 1.02 to 1.21; $I^2 = 67\%$). PP analysis reported similar results (Table 2, Supplement Figures S5–S6).

Subgroup analysis for H. pylori eradication rate by probiotic types

In ITT analysis, the combination of probiotics (8 studies) showed a higher eradication rate in the pretreatment group (RR = 1.15, 95% CI: 1.07–1.23; $I^2 = 20\%$), while a certain probiotic (4 studies) also demonstrated higher rates (RR = 1.12, 95% CI: 1.05–1.20; $I^2 = 63\%$). In PP analysis, the combination of probiotics yielded similar results (RR = 1.12, 95% CI: 1.05–1.18; $I^2 = 53\%$), whereas the certain probiotic showed comparable rates (RR = 1.13, 95% CI: 1.07–1.20; $I^2 = 73\%$). Both probiotic types were effective (Table 2, Supplement Figures S7–S8).

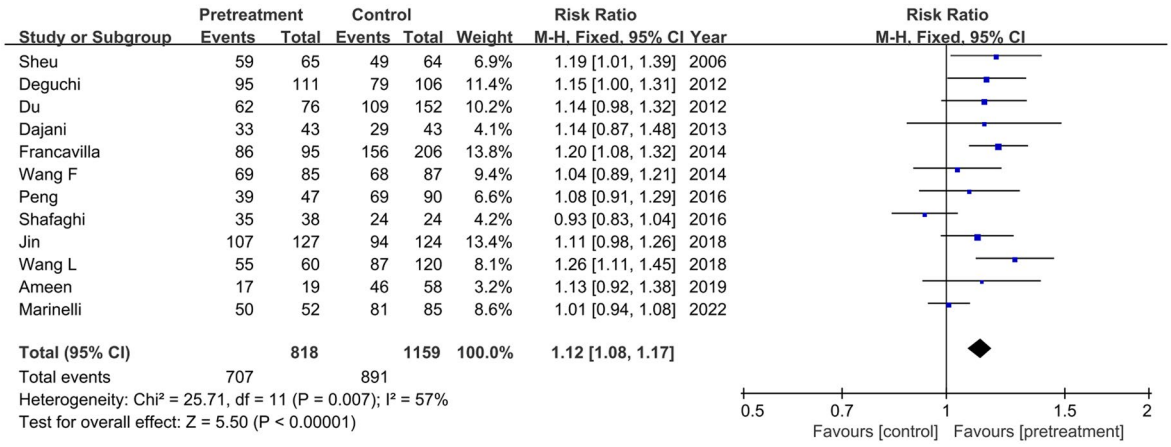


Figure 3. Forest Plot of randomized controlled trials of *H. pylori* eradication therapy: Efficacy on *H. pylori* eradication rate (per-protocol analysis).

Subgroup analysis for H. pylori eradication rate by probiotics pretreatment duration

In ITT analysis, the duration of probiotics pretreatment prior to *H. pylori* eradication therapy yielded inconsistent results. Pretreatment for 3 to 7 days did not significantly improve the eradication rate (RR = 1.06, 95% CI 0.93 to 1.22, $p = 0.39$, I^2 = 89%). However, pretreatment for 14 days significantly enhanced the eradication rate (RR = 1.14, 95% CI 1.08 to 1.21, $p < 0.001$, I^2 = 5%). Similarly, pretreatment for 21 days or longer also improved the eradication rate (RR = 1.16, 95% CI 1.05 to 1.28, $p = 0.003$, I^2 = 0%). PP analysis reported similar results. (Table 2, Supplement Figures S9–S10).

Subgroup analysis for H. pylori eradication rate by study design

In ITT analysis, significant improvements were observed in *pre vs. free* (RR = 1.21, 95% CI 1.12 to 1.31, $p < 0.001$, I^2 = 25%) and *pre and combine vs. free* (RR = 1.23, 95% CI: 1.12–1.36; I^2 = 0%), while other study designs with fewer studies did not show significant effects. In PP analysis, *pre vs. free* (RR = 1.20, 95% CI: 1.11 to1.29; I^2 = 16%) and *pre and combine vs. free* (RR = 1.21, 95% CI: 1.11 to1.32; I^2 = 0%) remained significant, with other designs yielding non-significant findings. Pretreatment before *H. pylori* eradication therapy showed consistent efficacy, particularly in comparison to free regimens (Table 2, Supplement Figures S11–S12).

Side effects

The main reported side effects included nausea, diarrhea, constipation and metallic taste et al. Some studies reported that side effects were similar, and some studies did not report side effects. According to available data from eight studies (1551 patients), side effects were less likely in the probiotics pretreatment group compared to the control group (16.0% vs. 28.3%, RR = 0.59, 95% CI 0.41 to 0.84, $p < 0.005$, Figure 4).

Risk of bias and sensitivity analysis

The risk of bias summary was shown in Figure 5. The risks for bias were shown in Supplement Figure S13. In the sensitivity analysis, after deleting one study at a time, the statistical significance of the combined RR remained unchanged.

Publication bias

Funnel plot showed no significant publication bias among the twelve studies included in this meta-analysis (Supplement Figure S14).

Table 2. Summary of Forest plot for subgroup analysis.

Sub-group		No. of studies	Intention-to-treat analysis						Per-protocol analysis					
			Pretreatment		Control		Risk Ratio		Pretreatment		Control		Risk Ratio	
			Events	Total	Events	Total	MH, Fixed,95%CI	I^2 Test	Events	Total	Events	Total	MH, Fixed,95%CI	I^2 Test
Country	studies from China	6	391	485	476	671	**1.14 [1.07, 1.22]**	0%	391	460	476	637	**1.13 [1.07, 1.20]**	0%
	studies from outside China	6	316	395	415	593	**1.13 [1.05, 1.22]**	60%	316	358	415	522	**1.11 [1.05, 1.18]**	74%
Regimen of eradication	STT	6	332	417	488	709	**1.15 [1.07, 1.23]**	0%	332	391	488	655	**1.15 [1.08, 1.22]**	0%
	BQT	6	375	463	403	555	**1.13 [1.05, 1.21]**	66%	375	427	403	504	**1.10 [1.04, 1.16]**	75%
H. pylori group	initial	2	157	193	188	270	**1.17[1.05, 1.30]**	0%	157	187	188	258	**1.14[1.04, 1.26]**	0%
	rescue	5	329	407	367	515	**1.14[1.06, 1.23]**	12%	329	384	367	485	**1.13[1.06, 1.21]**	12%
	unclear	5	221	280	336	479	**1.11[1.02, 1.21]**	67%	221	247	336	416	**1.10[1.04. 1.17]**	77%
Probiotic type	combination of probiotics	8	414	526	466	690	**1.15 [1.07, 1.23]**	20%	414	484	466	610	**1.12 [1.05, 1.18]**	53%
	certain probiotic	4	293	354	425	574	**1.12 [1.05, 1.20]**	63%	293	334	425	549	**1.13 [1.07, 1.20]**	73%
Probiotics pretreatment duration	3d-7d	2	85	104	105	136	1.06 [0.93, 1.22]	89%	85	90	105	109	0.98 [0.92, 1.04]	32%
	14d	7	387	477	584	830	**1.14 [1.08, 1.21]**	5%	387	447	584	777	**1.15 [1.09, 1.22]**	0%
	>=21d	3	235	299	202	298	**1.16 [1.05, 1.28]**	0%	235	281	202	273	**1.13 [1.04, 1.23]**	0%
Study design	pre vs. free	7	370	459	306	460	**1.21[1.12, 1.31]**	25%	370	439	306	436	**1.20[1.11, 1.29]**	16%
	pre vs. combine	3	147	178	138	178	1.07[0.96, 1.18]	0%	147	170	138	171	1.07[0.97, 1.18]	0%
	pre vs. after	1	62	78	61	77	1.00[0.85, 1.18]	/	62	76	61	74	0.99[0.85, 1.15]	/
	pre+combine vs. free	4	252	317	209	321	**1.23[1.12, 1.36]**	0%	252	289	209	290	**1.21[1.11, 1.32]**	0%
	pre+combine vs. combine	4	192	240	175	246	**1.13[1.02, 1.24]**	36%	192	216	175	210	1.06[0.98, 1.14]	48%
	pre+combine vs. after	1	17	37	15	24	0.74[0.46, 1.17]	/	17	19	15	18	1.07[0.83, 1.39]	/
	pre+combine+after vs. free	1	50	66	81	98	0.92[0.78, 1.08]	/	50	52	81	85	1.01[0.94, 1.08]	/

BQT: bismuth quadruple therapy, pre: pretreatment, STT: standard triple therapy.

Quality of evidence through GRADEPro analysis

The quality of evidence from the included studies was assessed using the Cochrane review tool GRADEpro GDT (https://gdt.gradepro.org/app/), focusing on the efficacy of probiotics pretreatment in *H. pylori* eradication therapy. As shown in Table 3, the certainty of the overall effects of probiotics pretreatment ($n = 12$) was high (⊕⊕⊕⊕) in the ITT analysis, demonstrating the effectiveness of probiotics in improving *H. pylori* eradication rates. However, the certainty decreased from high to moderate in PP analysis. Across all subgroup analyses, the certainty of evidence for probiotics pretreatment remained either high or moderate. Detailed GRADEpro assessments for all subgroups are provided in Supplement (Supplement Table S1).

4. Discussion

This systematic review and meta-analysis investigated the efficacy of probiotics pretreatment in improving *H. pylori* eradication rate. Unlike previous studies, which primarily focused on probiotics administered during or after *H. pylori* eradication therapies, this study explored the role of probiotics as a pretreatment strategy. Probiotics pretreatment significantly increased eradication rate (80.34% vs. 70.49%, RR = 1.14, 95% CI 1.08–1.19, $p < 0.001$). Subgroup analyses confirmed benefits across regions and eradication regimens, *H. pylori* group, with both single and combined probiotics showing efficacy. However, pretreatment duration was critical: 14 days or longer improved eradication rates, while shorter durations (3–7 days) did not. Study designs *pre vs. free* and *pre and combine vs. free* further validated probiotic pretreatment's efficacy in improving *H. pylori* eradication rate. These results suggest that probiotics, when administered prior to *H. pylori* eradication therapy, may play a beneficial role in improving treatment outcomes.

Two previous meta-analyses examined the efficacy of probiotics administered at different timings (before, during, after) in *H. pylori* eradication therapy. Lv et al. found that probiotics used before and after the treatment regimen improved eradication rates [40], while Shi et al. reported that probiotics administered before and throughout eradication therapy enhanced efficacy [10]. These meta-analyses included 4 or 5 RCTs with probiotics used before *H. pylori* eradication, and 1or 2 RCTs did not meet the current study's eligibility criteria; and their findings align with our study results. These findings contrast with earlier research, which suggested that while certain probiotics, such as *Lactobacillus*, can inhibit *H. pylori* growth, adhesion, and colonization through mechanisms like autolysin secretion and reduced binding to glycolipid receptors,[16-18] they do not clear *H. pylori* infection on their own [41]. For instance, single probiotic strains have been shown to reduce urease activity in *H. pylori*-infected patients but fail to eradicate the pathogen [21,42].

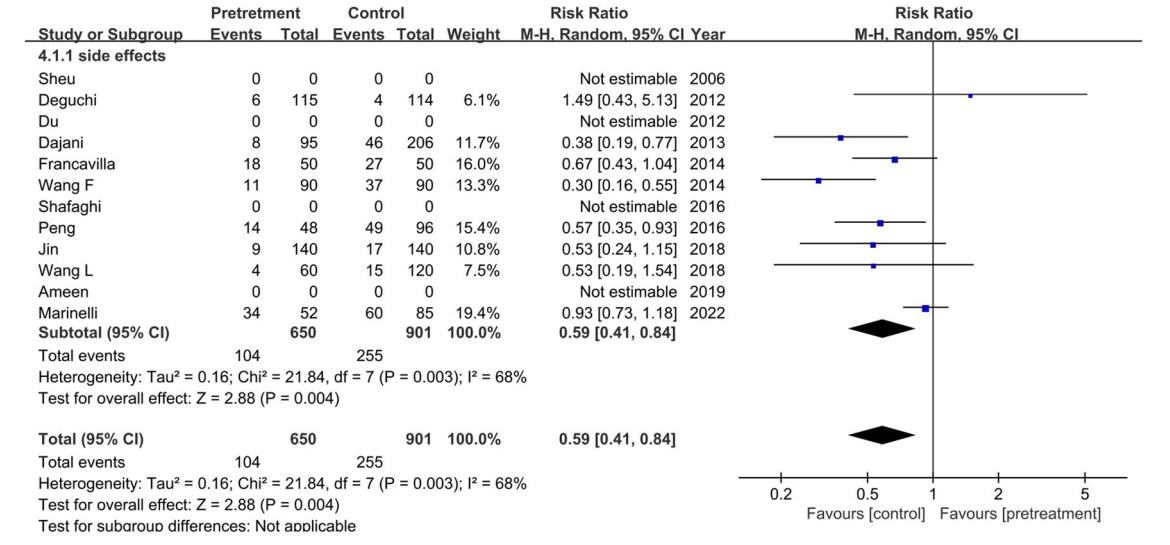


Figure 4. Forest Plot of randomized controlled trials of *H. pylori* eradication therapy: side effects.

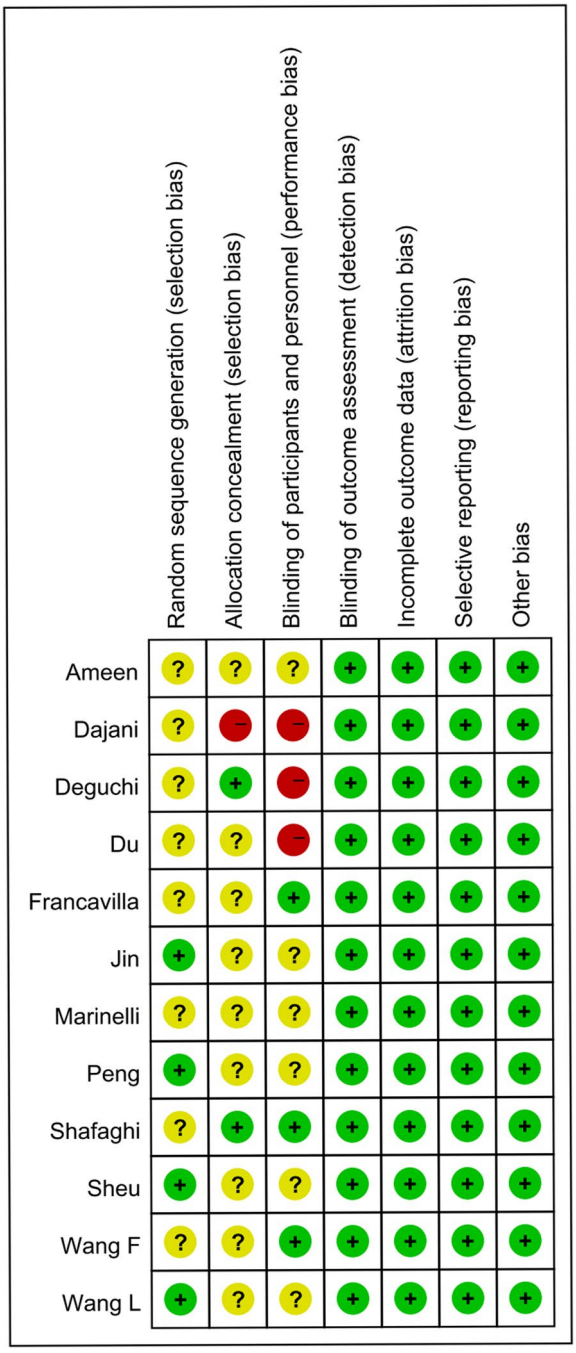


Figure 5. Risk of bias summary.

The potential mechanisms underlying the benefits of probiotics pretreatment may involve biofilm formation and gut microbiome modulation. Some probiotics can bind to and produce biofilms in cultured cells, significantly reducing *H. pylori* binding and colonization [43,44]. Additionally, supplementing with probiotics alters the gut microbiome, which may create an environment less conducive to *H. pylori* survival [19–21]. This is supported by studies showing that FMT cleared *H. pylori* in 40.6% of patients, highlighting the gut microbiome's potential role in *H. pylori* treatment [24]. Furthermore, a meta-analysis involving patients with inflammatory bowel disease (IBD), a condition associated with significant gut microbiome changes, found a reduced *H. pylori* infection rate in IBD patients, further emphasizing the interplay between the gut microbiome and *H. pylori* clearance [45].

The gut microbiome is increasingly recognized as a critical factor in maintaining intestinal homeostasis and immune function, offering hope for the treatment of infectious diseases. For example, FMT has proven

Table 3. GRADEpro Summary of findings for assessing the quality of evidence as assessed separately by two reviewers.

Certainty assessment							№ of patients		Effect			
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Pretreatment of probiotics	control	Relative (95% CI)	Absolute (95% CI)	Certainty	Importance
Total: intention-to-treat analysis (ITT)												
12	randomised trials	not serious	not serious	not serious	not serious	none	707/880 (80.3%)	891/1264 (70.5%)	**RR 1.14** (1.08 to 1.19)	**99 more per 1,000** (from 56 more to 134 more)	⊕⊕⊕⊕ High	CRITICAL
Total: per-protocol analysis (PP)												
12	randomised trials	not serious	serious	not serious	not serious	none	707/818 (86.4%)	891/1159 (76.9%)	**RR 1.12** (1.12 to 117.00)	**92 more per 1,000** (from 92 more to 1,000 more)	⊕⊕⊕○ Moderate	CRITICAL

CI: confidence interval; **RR:** risk ratio.

effective against *Clostridium difficile* infection and shows promise in treating human immunodeficiency virus infection [46–48]. Similarly, probiotic supplementation has been associated with reduced mortality in COVID-19 patients and may help decolonize pathogenic bacteria from the gut [49]. These findings, along with our results, suggest that modulating the gut microbiome—whether through probiotics, FMT, or other methods—could be a valuable strategy for addressing multidrug-resistant *H. pylori* and the long-term resistant bacterial communities that arise from eradication therapies. However, the mechanisms underlying these effects, such as the migration of gut-licensed immune cells to other tissues, require further investigation [50].

Despite these promising findings, this study has several limitations. First, the meta-analysis included only twelve RCTs with varying probiotic strains, doses, durations, and combinations, which may affect the reliability of the conclusions. Second, six of the included studies were conducted in China, and although studies from other regions supported the same conclusions, more diverse populations are needed to generalize the findings. Third, this study did not evaluate the simultaneous use of probiotics during eradication therapies or their supplementation after treatment, as these topics have been extensively covered in previous umbrella meta-analyses [9]. Future research should investigate key factors affecting probiotic efficacy, including strain selection, dosage optimization, antimicrobial interactions, and ethnic/age-specific responses through rigorous clinical trials and omics technologies.

In conclusion, this meta-analysis provides evidence that probiotics pretreatment significantly improves *H. pylori* eradication rate and reduces side effects, particularly when probiotics is used for an adequate duration. These findings highlight the potential value of gut microbiome modulation in *H. pylori* treatment and offer a new direction for addressing the challenges of antibiotic resistance and treatment failure. Further studies are needed to elucidate the mechanisms involved and to develop standardized protocols for probiotic use in *H. pylori* eradication therapies.

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Ethics statement

This meta-analysis is based on existing published RCTs. The Ethics Committee of Sichuan Academy of Science & Sichuan Provincial People's Hospital hereby declares that this meta - analysis is exempt from ethical review.

Author contributions

CRediT: **Yinghui Zhang**: Investigation, Methodology, Validation, Visualization, Writing – original draft; **Mengtian Tu**: Data curation, Formal analysis, Investigation, Software, Writing – original draft; **Ping Long**: Data curation; **Jing Zheng**: Data curation; **Guanghong Du**: Data curation; **Shiyu Xiao**: Investigation, Methodology, Software, Visualization, Writing – review & editing; **Caiping Gao**: Conceptualization, Data curation, Funding acquisition, Project administration, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data will be made available from the corresponding author [Caiping Gao] on reasonable request.

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